KCE PATHWAYS · GRADES 4–5 · UNIT 4: SHAPES AND SPACE · WEEK 16 · MATH

Apple Harvest in the Blue Ridge

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Fair Shares at the Orchard

Silas is helping at his family's apple orchard in the Blue Ridge Mountains. Warm up before the lesson, and show your work. Some problems may feel new, and that is okay!

1. Review (Unit 3): Multiply. Write each product in simplest form. a) 3/4 × 8 b) 2/5 × 1/3

2. Ms. Dunbar has 1/2 of a pan of apple bread left. She cuts it into 2 equal pieces. Draw the WHOLE pan and shade the half. Then split the half into 2 equal pieces. What fraction of the whole pan is each piece?

3. Mr. Barrera has 4 pounds of dried apple chips. Each bag holds 1/2 pound. Draw a number line from 0 to 4 and mark jumps of 1/2. How many bags can he fill?

4. Silas shares 1/3 of a pie equally with 1 friend, so 2 people share it. Each share is...

   bigger than 1/3 | smaller than 1/3 | exactly 1/3

5. In the division problem 12 ÷ 4 = 3, which number is the quotient (the answer)?

   12 | 4 | 3

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Apple Harvest in the Blue Ridge

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: Orchard Problems

Write an equation for each problem, solve it, and show your work.

1. Ms. Dunbar has 1/4 of a pan of apple crisp left. She cuts it into 3 equal pieces. Two pickers each eat 1 piece. What fraction of the WHOLE pan did the two pickers eat in all? Write it in simplest form.

2. Mr. Barrera packs 6 pounds of dried apple chips into 1/3-pound bags. He sells 10 bags at the morning market. How many bags are left?

3. Which equation matches this story? "Grandma Lucinda has 1/2 gallon of apple butter. She splits it equally into 5 jars. How much of a gallon is in each jar?"

   5 ÷ 1/2 = 10 | 1/2 ÷ 5 = 1/10 | 1/2 × 5 = 2 1/2 | 5 ÷ 2 = 2 1/2

4. Find the mistake: Delia writes 1/6 ÷ 2 = 1/3. In two or three complete sentences, explain what she did wrong, write the correct quotient, and check it with multiplication.

★ **Challenge:** Silas has 1/3 of an hour to wash 4 cider jugs, with equal time for each. How many MINUTES does each jug get? How many minutes would 6 jugs take at that pace?

Looking Ahead: Week 17: dividing decimals at a farmers market in Honolulu, Hawai'i, with money and metric units.

KCE PATHWAYS · GRADES 4–5 · UNIT 4: SHAPES AND SPACE · WEEK 16 · READING

The Storm Watchers of Norman

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Clues Inside Words

Wyatt and Noor help a retired weather scientist run a neighborhood weather station in Norman, Oklahoma. Show what you already know. It is fine to guess!

1. Review (Unit 3): Rewrite the sentence with quotation marks, a comma, and capital letters where they belong: noor said the wind is getting stronger

2. The word thermometer has two parts: therm and meter. What do you think each part means?

3. In the word unsafe, the prefix un- means...

   not | again | before | full of

4. Read: "The puppy was timid, so it hid under the porch whenever a stranger came near." What does timid most likely mean? Which words helped you?

5. Write two words you know that begin with pre- or re-. Tell what each word means.

KCE PATHWAYS · GRADES 4–5 · UNIT 4: SHAPES AND SPACE · WEEK 16 · READING

The Storm Watchers of Norman

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Word Detective Report

Use roots, affixes, context clues, and spelling rules, and write like an editor.

1. Break each word into its parts. Write what each part means and what the whole word means. a) invisible b) portable c) geologist

2. Spell each new word. a) un + rely + able b) dis + agree + ment c) the "not" prefix + regular

3. Be the editor. Rewrite each sentence correctly. a) after the storm passed noor said the barometer is rising again b) Ms. Nwosu showed Wyatt and I the rain gauge.

4. Writing: Write 3 or 4 sentences about a storm you remember. Use two words from today's lesson (such as predict or visible) and one context clue. Check capitals, commas, and end marks.

★ **Challenge:** Invent a name for a new weather tool from two word parts in the lesson. Write one sentence that defines it.

Looking Ahead: Week 17: reading a play set on a Florida strawberry farm, with scenes, stage directions, and dialogue.

KCE PATHWAYS · GRADES 4–5 · UNIT 4: SHAPES AND SPACE · WEEK 16 · SCIENCE

Waves, Light and How We See

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. **Review (Unit 3):** Name one energy change you learned about in Week 12, such as in a flashlight or a solar panel. Write the form of energy before and after.

2. Drop a pebble into a pond and rings of ripples spread out. What do you think is moving outward: the water itself, or something else? Explain your guess.

3. Draw a wave. Label where you think its highest point and lowest point are.

4. How do you think your eyes see a book on a table in a lit room? Draw arrows to show which way you think the light travels.

5. Name one way people send messages over long distances without talking face to face. How does it work?

KCE PATHWAYS · GRADES 4–5 · UNIT 4: SHAPES AND SPACE · WEEK 16 · SCIENCE

Waves, Light and How We See

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Show What I Learned

Use evidence from today's lesson. Write in complete sentences.

1. Draw a wave with a dashed rest line. Label the **crest**, **trough**, **amplitude** and **wavelength**.

2. **Two steps:** A wave drawing has crests at squares 3, 11 and 19, and each crest rises 4 squares above the rest line. (a) What is the wavelength? (b) What is the amplitude?

3. **Find the mistake:** "I can see my red shirt in a totally dark closet, because it is red." Explain what is wrong. Use the word **reflect**. Then name one thing that **refracts** light.

4. **Mirror Lab:** A beam hits a mirror at 25 degrees from the normal. Predict the reflected angle. Then explain why one trial that measured 60 and 58 degrees does not prove the law of reflection wrong.

★ **Challenge:** Use these Morse letters: A = · –, E = ·, H = · · · ·, I = · ·, N = – ·, O = – – –, S = · · ·, T = –. Write a 3-letter word in dots and dashes. Why are two kinds of signals hard to misread?

Looking Ahead: Next week: **Energy Resources.** Compare fossil fuels, nuclear, solar, wind, water, geothermal and biomass energy with real U.S. Energy Information Administration data. Bring this question: where does your electricity come from?

KCE PATHWAYS · GRADES 4–5 · UNIT 4: SHAPES AND SPACE · WEEK 16 · SOCIAL STUDIES

Railroads, Factories and New Americans (1870–1915)

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Who Did the Work?

Write what you know right now. Guesses are fine; this is your starting point.

1. **Review (Unit 3):** Who did most of the work on the Central Pacific Railroad, and what did those workers go on strike for in 1867?

2. What do you already know about immigrants who came to the United States around the year 1900? Where did they come from, and why?

3. Guess: Why do you think some children around 1900 worked in mills, mines and city streets instead of going to school?

4. When workers go on **strike**, what do you think they do? Why might they do it?

5. Write one question you have about factories, immigrants or workers long ago.

KCE PATHWAYS · GRADES 4–5 · UNIT 4: SHAPES AND SPACE · WEEK 16 · SOCIAL STUDIES

Railroads, Factories and New Americans (1870–1915)

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Who Did the Work?

Use evidence from today's lesson. Write in complete sentences.

1. Choose one Lewis Hine caption from today. What facts did he write down, and why?

2. **Find the mistake:** "Ellis Island and Angel Island treated every immigrant the same way." Explain what is wrong, using evidence.

3. **Two steps:** The census counted 1,990,225 working children ages 10 to 15 in 1910 and 1,060,858 in 1920. (a) How many fewer is that? (b) Why might the drop look bigger than it really was?

4. Andrew Carnegie paid for libraries and also backed breaking the Homestead union. Write two fair sentences about him.

★ **Challenge:** You are a garment worker at Cooper Union on November 22, 1909. Write two sentences to the crowd about why you will strike, using one fact from the lesson.

Looking Ahead: Next week: Jim Crow and the fight for equal rights, from Plessy v. Ferguson to Ida B. Wells.